

Optimization of the Resources Management in Fighting Wildfires

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ABSTRACT / Wildfires lead to important economic, social, and environmental losses, especially in areas of Mediterra-

nean climate where they are of a high intensity and frequency. Over the past 30 years there has been a dramatic surge in the development and use of fire spread models. However, given the chaotic nature of environmental systems, it is very difficult to develop real-time fire-extinguishing models. This article proposes a method of optimizing the performance of wildfire fighting resources such that losses are kept to a minimum. The optimization procedure includes discrete simulation algorithms and Bayesian optimization methods for discrete and continuous problems (simulated annealing and Bayesian global optimization). Fast calculus algorithms are applied to provide optimization outcomes in short periods of time such that the predictions of the model and the real behavior of the fire, combat resources, and meteorological conditions are similar. In addition, adaptive algorithms take into account the chaotic behavior of wildfire so that the system can be updated with data corresponding to the real situation to obtain a new optimum solution. The application of this method to the Northwest Forest of Madrid (Spain) is also described. This application allowed us to check that it is a helpful tool in the decision-making process.

In a system scarcely affected by man's action, wildland fires may be considered natural processes, which favor the development of ecosystems. However, natural fires can occasionally lead to catastrophic situations, which require direct fire-fighting action. This requirement becomes more evident in a highly populated system where property and human lives are at risk. There is an important international concern about the wildfire problem. Present-day European forestry policy criteria includes guidelines to prevent, control, and minimize the damage produced by wildfires (Annex 1 of the Resolution L2 Pan-European Criteria and Indicators for Sustainable Forest Management). Every year, there are about 45,000 wildfires in Europe. Between 1989 and 1993 alone, 2.6 million hectares were de-

stroyed. This is the same as if every year an area equal to the surface of Belgium were burnt. The consequences of wildfires in Spain are very serious, and the annual average losses often are higher than 1,500 million dollars. In the year 1994 only, there was a 1.64% loss of existing forest vegetation (Vélez 2000b). Besides taking silvicultural actions in autumn and winter to minimize the consequences of fires, it is necessary to develop tools that may serve to provide action guidelines in real time to minimize the effects of future fires.

The consequences of a wildfire may be predicted using models that simulate its spread and take into account the spatio-temporal action of fire-fighters. These models help combat wildfires, and also enable simulations of wildfires, which helps forestry management decision-making.

The aim of the present paper is to present an optimization model for resource management that minimizes the value of the burnt surface, using simulation models that predict the behavior of a fire and the resources' combat. Deterministic models are usually utilized to predict the spread of the fire (Dupuy 1991).

KEY WORDS: Bayesian methods; Mockus adaptive model; Simulated annealing; Expansion models; Wildfire management

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Among them, the most widely used BEHAVE (Andrews 1986), that applies the model developed by Rothermel and Burgan (1984). Rothermel's first studies, dated in 1960, consisted in the study of fire spread in a laboratory, assuming fuel to be homogeneous and a constant wind speed. Albini (1986) further applied Rothermel's model to non-stationary wind conditions and assumed that fire expands in an elliptical way. Anderson (1989) developed a fuel modeling system and introduced the width coefficient and the flame height in BEHAVE. In the present investigation, the CARDIN (Martínez Millán and others 1994) model was adopted to estimate the spread of the fire. CARDIN uses raster maps and applies BEHAVE algorithms in a discreet way to each pixel of the area under study. In CARDIN, when weather conditions, ground slope, and fuel load are constant, fire expansion is assumed to occur in a cardioide way, instead of assuming an elliptical expansion, as it is done in BEHAVE (Merida 2000). The spatio-temporal allocation of the resources at the start of fire-fighting represented the decision variables.

Concerning optimization, Wybo's (1992) methods, are based on the construction of expert systems. However, we chose to develop Bayesian optimization procedures due to the requirements of our particular problem—the need to include subjective information and the consequences of the adopted decision—and in particular, due to its applicability to relatively frequent small or medium-sized fires in which it is necessary to define general plans of action. Environmental variables are generally chaotic in that small modifications in the starting variables may lead to disproportionate effects. Previous applications of CARDIN's model in Spain have proved that this model is accurate for time intervals of 20 min, i.e., it is necessary to update the system every 20 min, given the small discrepancies that emerge between the information initially introduced in the fuel maps, ground slope values, and weather parameters, and those corresponding to reality. To solve this problem, the optimization procedure developed is adaptive and adaptation times shorter than 15 to 30 min are foreseen. It is supposed that during this period of time the discrepancy between the results of the model and reality will not affect the outcome of the fire.

Methodology

The Problem

The value of the affected or burnt-surface area may be considered as a scalar function of the vectorial variable, which includes variables that can and cannot be

controlled by the manager. The uncontrollable variables are:

- Wind speed and direction
- Fuel type map
- Territory digital model
- Relative humidity of dead fuel—dependent on the time of the day when the fire starts, of the air relative humidity, of temperature and altitude
- Relative humidity of live fuel
- The unitary performance of each type of combat resource
- The time when the resources arrive at the fire

The controllable variables are:

- Type and number of resources
- Initial allocation of the combat resources, from where fire starts to be fought

The best combat strategy (to optimize the fire-fighting process), minimizing the deleterious effects of the fire, needs to be defined using the available resources and information. We calculated the burnt-surface value by means of data integration models. These allow one integrated value to be assigned to each point of the territory. This value represents the "overall merit," of the point analyzed. The higher this value is, the higher priority for fire defense. The value function of point f is made up of economic, environmental, and social factors and requires the application of scaling techniques (Martínez Falero and others 1995a).

The overall merit function of the surface is the value function or the objective of the optimization problem. If Ω is the domain or the geographical area under investigation, the objective function is given by:

$$F_{\Omega}(t) = \int_{\Omega} f(x, y, t) d\omega$$

and is dependent upon point location as well as time as a consequence of the damage caused by the fire.

In practice, function f is unknown. What is known is a discrete map of mean values on a lattice Γ . Generally, this lattice is rectangular or square and has N_y rows and N_x columns. The overall merit function of the study surface Ω approximates

$$F_{\Omega}(t) \approx \overline{F_{\Omega}}(t) = \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \overline{f_{ij}}(t),$$

where $\overline{f_{ij}}(t)$ is a mean value which represents the overall merit function in the pixel associated with the indices

(discrete coordinates) (ij) , at time t after the start of the fire.

The objective function mainly depends upon:

1. The spread of the fire represented by the state variable $\varphi(x, y, t)$ which has the following values: 1 if point (x, y) is burnt at time t ; 0 if point (x, y) is not burnt at time t .
2. The spread of the fire depends upon the decisional variable of resource management $\tilde{\theta}$. If m is the number of the fire-fighting resources available, $\tilde{\theta}(t) \in \mathfrak{R}^{4m}$ since, for each combat resource k , the following need to be established: t_k the time at which action is started, (x_k, y_k) the allocation where resource k starts its combat, and α_k the direction of action (clockwise or anticlockwise).

Now, we can formulate the a real overall merit function as

$$F_{\Omega}(\tilde{\theta}(t), t_f) = \int_{\Omega} f[\varphi(\tilde{\theta}(t), t_f)] d\omega$$

which is a scalar function of a vectorial variable of dimension $4m+1$, where t_f is the time of evaluation of the overall merit function, generally at the end of the fire.

In general, the overall merit of the area lost as a consequence of a fire is given by

$$\begin{aligned} \Delta F_{\Omega}(\tilde{\theta}(t), t_f) &= \int_{\Omega} (f(x, y, t_f) - f(x, y, 0)) d\omega \\ &= \int_{\Omega} (f(x, y, t_f) d\omega - F_0) \end{aligned}$$

which, in reality, represents the objective function.

Consequently, the problem that needs to be resolved is

$$\min \Delta F_{\Omega}(\tilde{\theta}(t), t_f), \quad \text{where} \quad \tilde{\theta} \in \Theta$$

where Θ is the set of possible resource-management strategies corresponding to the available resources.

If $\Delta F_{\Omega}(\tilde{\theta}_0(t), t_f) \leq \Delta F_{\Omega}(\tilde{\theta}(t), t_f) \forall \tilde{\theta} \in \Theta$ then $\tilde{\theta}_0$ is the optimum strategy looked for.

In practice, as it has explained before, the chaotic nature of our knowledge of uncontrollable variables means that determining an optimum strategy for the complete duration of the fire is not even considered. On the contrary, the choice of an optimum management strategy of fire-fighting resources corresponds to a limited time period δt . In real cases, δt represents a

period of 15–30 min. We propose Bayesian models for this reason.

Description of the Model

Let the time interval be (t_n, t_{n+1}) , then the wildfire spread and action of resource models assign a function, $\varphi_i(x, y, t)$, for the spread of the fire and a value of the objective function, $\Delta F_{\Omega, i}(\tilde{\theta}_{n, i}(t), t_f)$, to each strategy of management of the available resources, $\tilde{\theta}_{n, i}$, calculated for that specific time interval.

In order to abbreviate, here after $x_{n, i} = \tilde{\theta}_{n, i}$ where $X = \Theta$ and $y_{n, i} = \Delta F_{\Omega, i}[\tilde{\theta}_{n, i}(t), t_f]$.

The Bayesian procedures involve defining a decision rule ϑ over a set of samples Z , in which each $y_{i, n}$ is obtained by the application of simulation models,

$$Z = [(x_{n, 1}, y_{n, 1}), \dots, (x_{n, i}, y_{n, i}), \dots, (x_{n, p}, y_{n, p})],$$

where $y_{n, i}$ is the loss of overall value of the study area that occurs as a consequence of the fire and of the resource-management strategy adopted.

This decision rule ϑ will enable us to find the next value of the decisional variable x_{p+1} , closer to the optimum (Mockus 1989); $x_{n, p+1} = \vartheta(z)$.

Bayesian methods attempt to minimize mean losses or the decision risk which we shall represent as $\rho_{n, 0}(\vartheta)$, which, in general, takes the form:

$$\rho_{n, 0}(\vartheta) = y_{n, p+1}[x_{n, p+1}(\vartheta)] - \min_{x \in X} y_{n, p+1}[x(\vartheta)]$$

Thus, the only remaining decision is which decision rule enables us to approach the optimum. We have adopted Mockus' optimization method 1989 in which the function to be optimized is expressed as a combination of functions defined over connected intervals of the X variable,

$$f(x) = f_h(x) \quad \forall x \in [X_h, X_{h+1}]$$

where $h = 1 \dots n$

where h is the interval order number, whenever $\cup_{h=1}^n [X_h, X_{h+1}] = X$.

Mockus' simplification, limits the search for the next solution, x_{p+1} , to the boundaries of the contiguous intervals X_h . The value of x_h corresponding to the lowest y'_k determines the next sample value x_{p+1} .

This is an iterative process that converges to the optimum.

Reducing the size of the decisional variable x . The efficiency of the Mockus procedure depends upon the reduction in size of the decisional variable to obtain one unidimensional variable. Otherwise, the computer-response time would be too long because of the extremely large number of combinations resulting from

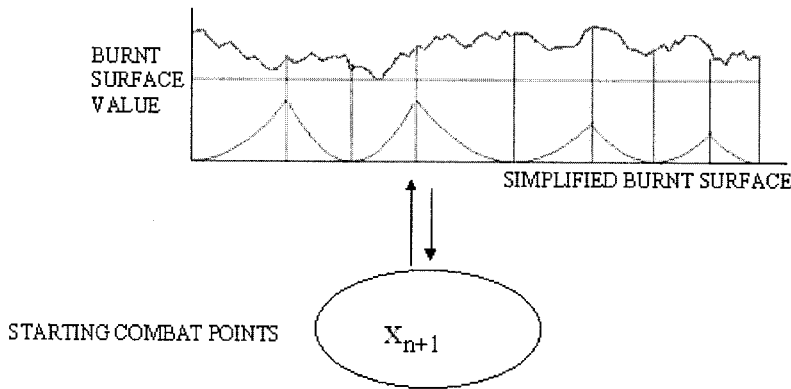


Figure 1. Relationship between the burnt-surface value and the simplified burnt surface (SBS).

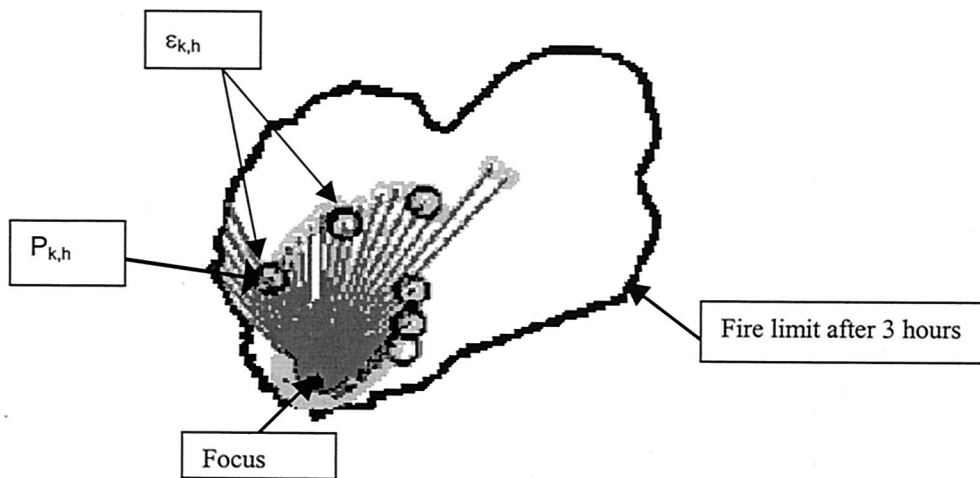


Figure 2. Starting combat point for a SBS, $P_{k,h}$ (point h in radius k). From this point, the resources perform an elliptical combat according to an ellipse $(\epsilon_{k,h})$.

assigning m resources of different types, arriving at each location at a different time, that can be fought in two different ways. Thus, the procedure would not be efficient. Therefore, we used the simplified burnt surface (SBS) variable, which will be defined later and is univocally determined using the decisional variable by the transformation, $S: \mathcal{R}^{4m} \rightarrow \mathcal{R}; x_k \rightarrow S_k$.

Figure 1 illustrates the relationship between the value of the burnt area (Y-axis), the decisional variable “points where combat starts,” and the new unidimensional variable SBS (X-axis). The graph also shows the connected intervals and the optimum value obtained after applying the Mockus algorithm

The SBS, see Figure 2, represents the area that would be burnt in a simplified combat characterized by the following:

1. The number of possible points of start of combat

are reduced to the points that belong to N radii computed from the focus. The resources cannot combat from those points where fire arrives before they do. The maximum length of each radius corresponds to the burnt distance after 3 h of fire. The time of 3 h is taken to represent a reasonable balance between the longest possible time which would lead to the most significant result possible, and the limitations of the model due to the imprecision and estimations involved in characterizing the environment in which the fire develops.

2. When the fire reaches the possible points of start of combat $P_{k,h}$, the burnt area is substituted by the Cullman ellipse (Martínez-Falero and others 1995b), defined so that its area is equivalent to the burnt area at a given time. The aim of the Cullman ellipse is to construct a flat, compact figure which is

GENERAL PROCESS

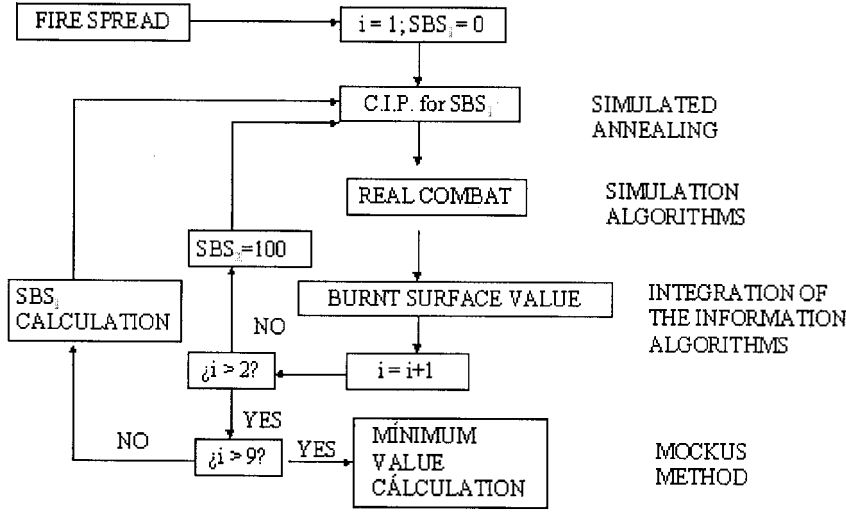


Figure 3. Process flow chart and related algorithms.

a good two-dimensional estimator of the surfaces generated during fires, preserving its main orientation and area. The focus of the fire is a focus of the ellipse. The second focus of the ellipse is the center of gravity of the points the fire reaches in a time under or equal to the time of arrival $t_{P_{k,h}}$ at the possible point of start of combat $P_{k,h}$. Where $P_{k,h} \in \epsilon_{k,h}$.

- From each possible point of start of combat $P_{k,h}$ (point h in radius k) the resources perform an elliptical combat according to a Cullman ellipse $\epsilon_{k,h}$. This kind of elliptical combat is only applied to calculate a simplified combat, and its aim is to calculate the allocation of the resources at the beginning of each iteration process. As will be seen later, for the calculation of the value of the burnt area, the combat calculation algorithm will resemble as close as possible a real combat
- For each fire-fighting resource, a map of resource arrival time from its initial allocation to the fire is obtained using another simulation model. Consequently, each pixel (ij) of the territory under study Ω is assigned a vector of arrival times of the different resources $\vec{T}_{ij}$. The combat of a resource c is performed in a clockwise or counter clockwise direction along the perimeter of the Cullman ellipse $\epsilon_{k,h}$, until fire conditions no longer allow combat. The SBS associated with the resource management strategy adopted will be determined by

$$S(x) = \frac{\pi}{N_r} \sum_{k=1}^{N_r} l_k^2$$

where l_k is the burnt length of the radius k .

However, the mapping of S is not one-to-one, thus, the transformation, S^{-1} needed to apply Mockus' method is not unique. Thus, given a particular SBS, the problem arises of finding S_0 , a value of the decisional variable x_0 , such that the associated SBS is as close to S_0 as possible.

$$|S(x_0) - S_0| \leq \min_{x \in X} |S(x) - S_0|$$

Fortunately, the simplifications described above permit this search to be approached as a combinatorial optimization problem.

The schema of the entire process is as follows, see Figure 3. In the following sections, the algorithms are described in detail.

- A map of fire arrival times is obtained using a spread model, in our case CARDIN, assuming no action of the combat resources, so that the system stores it in a file and does not have to recalculate it. The input information is the following: wind speed and direction, fuel type map, territory digital model, relative humidity of live fuel, relative humidity of dead fuel, which depends on the time of the day when the fire starts, air relative humidity, temperature, altitude, and the location where the fire starts. Each pixel will be assigned a fire arrival time depending on its features. Pixels ranging 20–50 m, are regarded as the most suitable for fire-spread simulations, because of existing cartography and the calculation speed of actual PCs (Martínez Millán and others 2000).

- The points of the boundaries of contiguous inter-

vals are calculated, applying the Mockus' method. First iteration: i. The first step is to obtain the starting combat points of the resources when the SBS is 0%. The simulated annealing method is applied in this step. ii. The following step consists of the calculation of the real burnt surface when the resources start their combat in the points obtained in the former step. iii. The value of the real burnt surface is calculated. Therefore, the first point of the first interval of Mockus is obtained. Second iteration: The starting combat points of the resources are obtained when the SBS is 100%. Then steps ii and iii are followed. Next iterations: The value of the next SBS is obtained applying Mockus's method. Then steps i, ii, and iii are calculated.

3. The sought starting combat points correspond to the ones of the burnt surface of minimum value.

Obtaining a value for the decisional variable from a simplified burnt surface value. Simulated annealing is applied as the combinatorial optimization method (Geman and Geman 1984, Cerny 1982, Kirpatrick and others 1982, etc.). This is an iterative method in which, starting from an initial SBS, this value is continuously changed until the looked-for SBS is reached.

For a given initial allocation of resources, x_{h-1} , corresponding to a particular value of the resource-management decisional variable, the probability that the resource k will change position from j to i , that the possible starting point of fire-fighting, c_k , with the initial positions of the other resources, remaining unaltered, $-c_k$ the allocation, x_k , follows a Gibbs distribution expressed by

$$P[x_{h,i} = c_k; \lambda(\xi_i) = \bar{c}_k] = \frac{1}{Z} \exp \left\{ -\frac{1}{T} \cup [x_{h,i} = c_k; \lambda(\xi_i) = \bar{c}_k] \right\}$$

if the direction of action on the corresponding Cullman ellipse is maintained, where Z is a standardization constant, T an adequate constant (in the original Gibbs model the *cooling temperature*), and $U[x_{h,i} = c_k; \lambda(\xi_i) = \bar{c}_k]$ is an energy function, a subjective measure of the proximity at the optimum of the value proposed.

The following energy function was adopted here:

$$U[x_{h,i} = c_k; \lambda(\xi_i) = \bar{c}_k] = |S_0 - S[x_{h,i} = c_k; \lambda(\xi_i) = \bar{c}_k]| \quad (12)$$

where $\lambda(\xi_i)$ is the allocation of resources at the remaining possible combat starting points, S_0 is the previously specified SBS, and $S[x_{h,i} = c_k; \lambda(\xi_i) = \bar{c}_k]$ is the SBS corresponding to a resource management decisional variable which comprises the allocation of resource k at

possible combat starting point i and the remaining resources continue to act as in the previous step.

At each step h of the reiterative process, the position of each available combat resource is individually changed at each of N possible starting points. The energy function and the probability of change are evaluated and the position of the resource yielding the greatest probability of change only is modified. The iterative process ends when the energy function becomes stable.

Simulation of the real combat of the fighting resources. Initially, the model determines the critical path between each fire-extinguishing resource allocation point and the point assigned as its combat starting point.

The combat path followed by each fire-extinguishing resource depends upon the type of fight performed at the point allocated to each resource at the start of the fight. In general, two types of combat may be defined: direct combat (extinguishing resources are placed at the perimeter of the fire and the combat path coincides with the fire front; for security reasons this type of combat is not generally performed on the maximally advanced fire front) and indirect combat (extinguishing resources make a firebreak in specific areas). The information needed for combat calculation comprises: the resources performance for each type of fuel and ground slope as well as for each type of resource, the maximum slope at which they can fight, the maximum combat time without a rest, the maximum flame length that allows direct combat, the location of water sources, the resources basis, a road map, a digital model of the area, the fuel map, and the fire spread map. The selection of the type of combat is done automatically based on the speed of the fire spread, flame height, and the type of crew available.

Direct combat (Figure 4) is performed manually by combat crews, by fire pumps, or by the action of planes or helicopters of known mean performance under diverse circumstances. Indirect attack is conducted by crews or by mechanical means using bulldozers, excavators, etc.

When a starting point permits direct combat, there is only one possible trajectory: the resources act on the fire following the front perimeter. Aerial vehicles are able to quickly unload water and position themselves over the fire-front perimeter for only an instant. In contrast, land-resource behavior is divided into constant time periods (minutes). The succession of connected points through which the combat path runs is calculated by computing the optimum fight each minute. Even the use of vehicles with water reservoirs, which require continuous filling, is univocal once the combat starting point has been defined. Once the res-

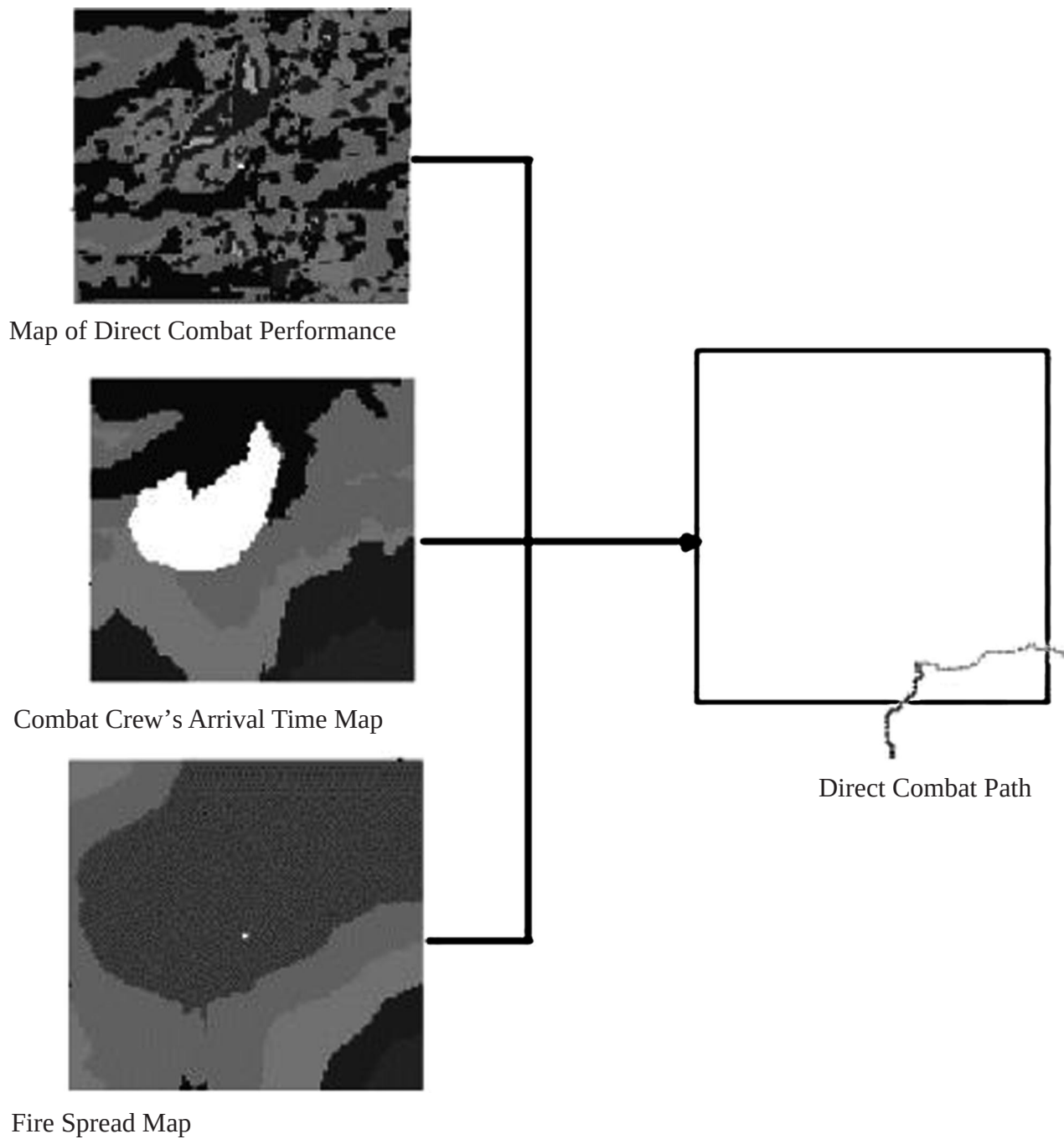


Figure 4. Computer simulation of a direct combat.

ervoir has been emptied, the vehicle must go to the nearest water-filling point and return to the starting point or to the closest point that the fire-front permits along critical paths.

A manual crew would have to stop combating because of high flame speed and large flame height, steep ground slope, or lack of working time. Aerial crews

would have to stop combating because of factors such as high wind speed, low visibility, or lack of combating time.

When indirect combat is undertaken (Figure 5), a firebreak's trajectory cannot be computed as a succession of optimum elementary firebreaks. Furthermore, there is more than one possible location for a firebreak

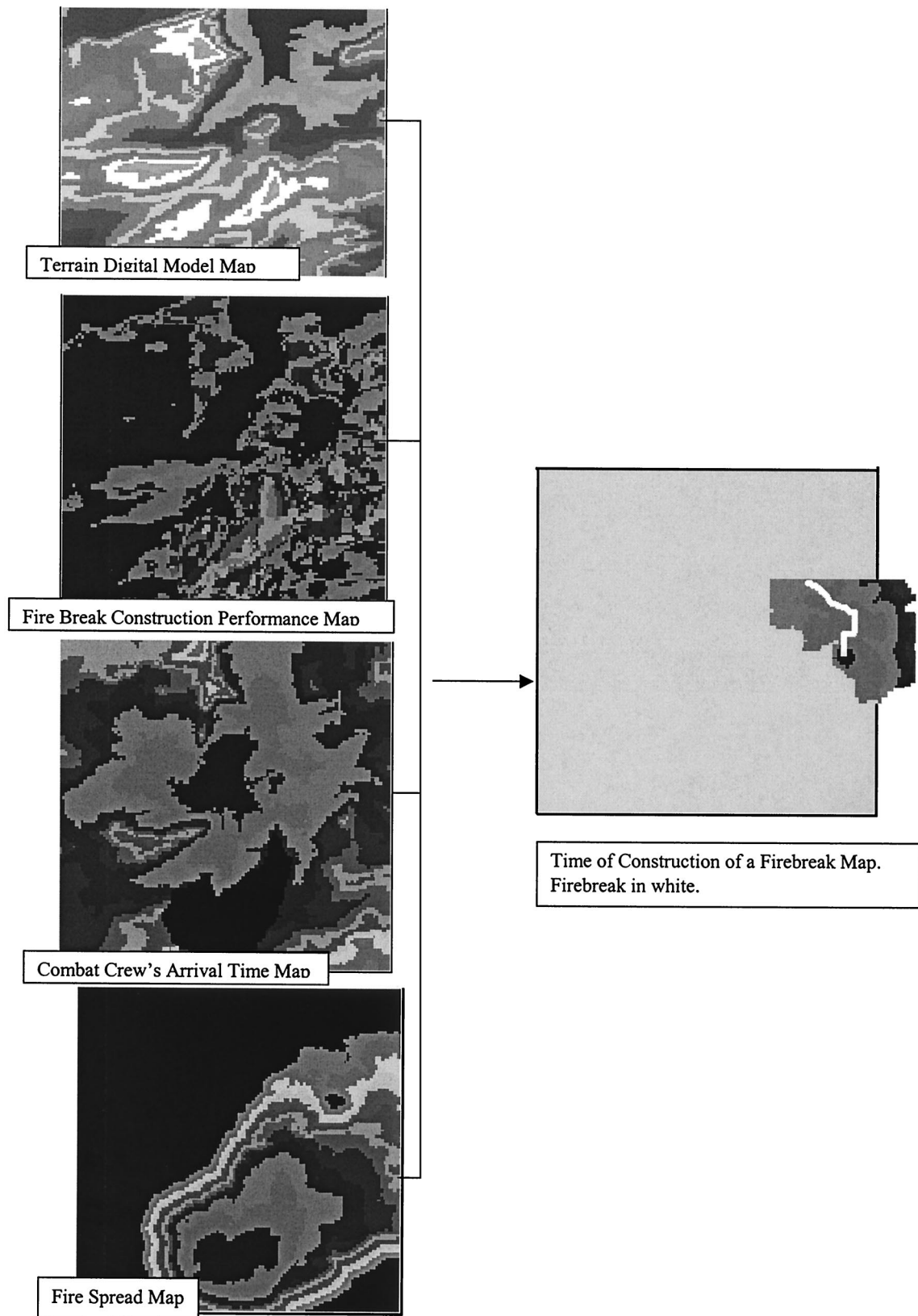


Figure 5. Computer simulation of an indirect combat.

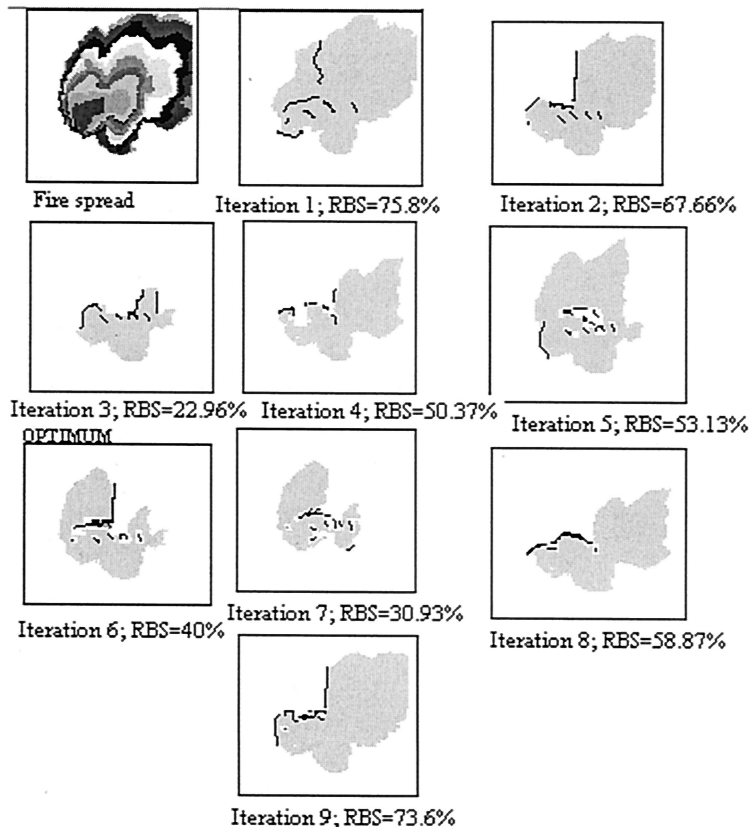


Figure 6. Results in the optimization iterative process. The surface can be seen with the combats performed by the resources. The best combat alternative is reached in the third iteration.

and its final position depends upon maintaining access to neighboring resources. A map showing the time of firebreak construction from the combat starting point to any other point in the territory is used whenever this time is less than the fire-arrival time. This map is obtained using the Dantzig's (1960) algorithm which identifies the shortest route from the combat starting point to the remaining area points where firebreaks can be constructed, building a minimum spanning tree, based on the mean performance of the resources according to local topographical conditions and types of fuel. The point of contact between the time map and the univocal path of the neighboring direct combat resources or over the time map of the neighboring indirect combat resources determines the end of the firebreak. The firebreak trajectory is given by the maximum slope curve (on the time map of the construction of firebreaks) between the intersection point and the indirect combat starting point.

A practical example. Next, a practical example is shown. The best combat is calculated in an area of the Guadarrama Mountains in Madrid. The conditions used in the optimization process are as follows: The study surface was 4 km². The wind speed was 10 km/h

the direction, west southwest. The temperature was 28°C, (corresponding to those prevailing during the day in the summer months according to data from the National Meteorological Service). The available resources considered for the simulation were: three fire fighting crews (manual); a CAT D-6 bulldozer; a 4,500-L fire pump and a 5,500-L fire-fighting plane. The maximum combat time was 6 h. The other information used of this area: Territory Digital Model, the fuel map, the road network, the map of the water points, map of the location of the resource basis, map of the protected areas, economic value of the study area and information about the performance of the resources under different conditions. The results obtained following the application of the optimization process are shown in Figure 6. Figure 7 shows the detail of the best combat alternative.

Application to the Forest of the Northwest of Madrid Region

The aim of this study was to determine the areas of public woodland within the Madrid province where the

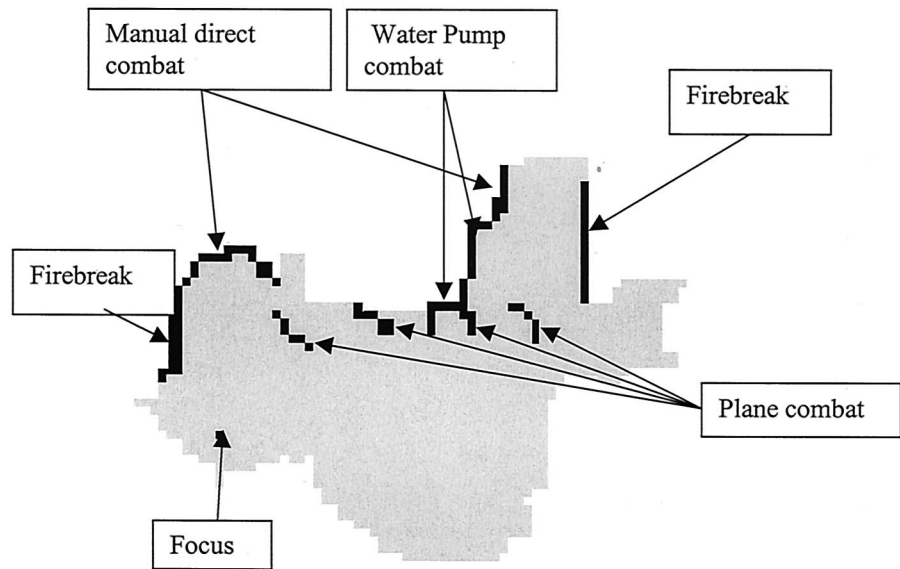


Figure 7. Detail of the result of an iteration of the optimization process. The direct and indirect combats performed by the resources are marked.

losses due to the wildfires were high when the meteorological conditions and the combat intensity were normal.

Figure 8 shows the study area and includes maps of soil use according to BEHAVE fuel classification models adapted by the *Instituto Nacional para la Conservación de la Naturaleza* (ICONA) (1987) to the types of fuel corresponding to the Madrid province. ICONA made a photographic characterization of fuel types for homogeneous forest regions of Madrid, using the 13 fuel models of Rothermel (Vélez 2000a). Also shown is the territory digital model. The pixel size corresponds to 25 m.

The procedure was as follows: The study points were randomly sampled. The size of the sample was 86 points. The surface of every point was 4 km². In each area studied, the optimum combat was calculated for three different wind directions and 10 different combat crew configurations, i.e., 30 different optimum combats were calculated in each area, giving a total of 2580 combats. With the aim of extrapolating these results to the remaining area of the Sierra de Guadarrama region, these combats were grouped according to their values of the variables “combat intensity” and “value of the burnt surface,” as will be explained later. Next, the territory features that could give rise to such specific groups were analyzed using an analysis of variance. Once the physical variables of each group were characterized, the remaining points of the territory were assigned to the group with more similar physical characteristics.

The best combat was calculated in all the points as it has been described above. This optimum combat was

calculated under different wind conditions and different combat intensity for every point.

Wind conditions. The wind conditions (corresponding to those prevailing during the summer months according to data from the National Meteorological Service) were: wind speed, 10 km/h; wind directions, west-southwest, north, south. The other meteorological conditions are the same as in *A practical example*.

Combat intensity. Types of resources available were: Manual resource; performs direct and indirect combat made up of six people. 5,500-L fire-fighting plane; performs direct combat. CAT D-6 bulldozer; performs indirect combat. 4,500-L fire pump; performs direct combat.

The best combat was calculated in each point for each wind speed with the resource configurations listed in Table 1.

Classification of the Results

The results of the simulations have been classified into three polythetic divisive classifications (Martín-Fernández 1998). This is a clustering technique whose aim is to classify a collection of units into groups, using an initial distance matrix. The starting point is the whole collection of units. This is then divided into two groups, that are more homogeneous than the original one. The dividing process is applied again to each of the resulting two groups, and the procedure is continued until a predetermined number of groups is reached, or until the groups obtained present an internal homogeneity considered sufficient for the purposes of the study. The polythetic method uses several variables in the classifi-

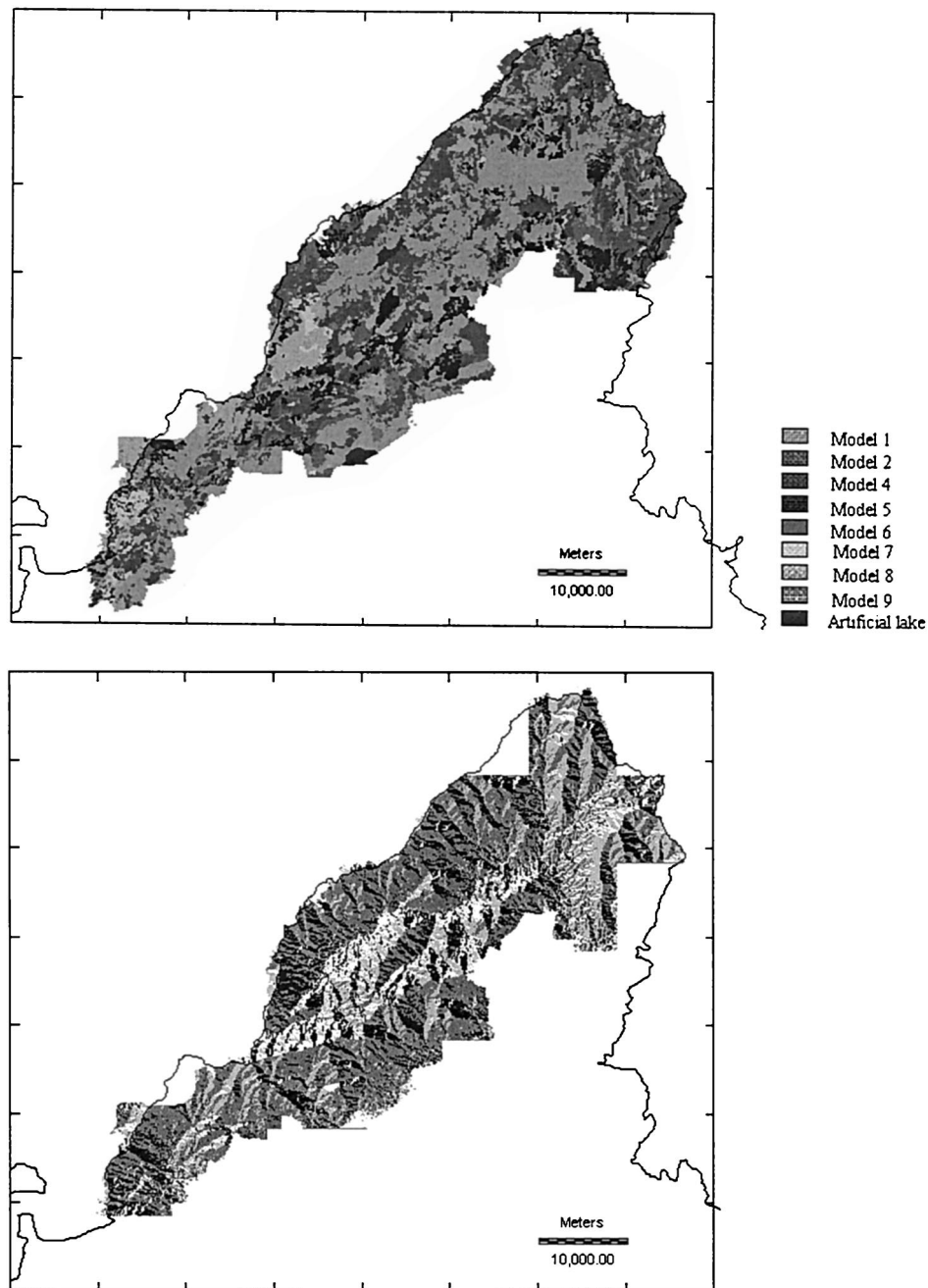


Figure 8. Study area: Fuel map of the study area, at the top of the figure, and territory digital model, at the bottom.

cation (Martínez-Falero and others 1995b). In the present case, this method has been applied to the results obtained for each wind type. The three classifications obtained are the following: Classification 1, West-South west wind, Classification 2, south wind, Classification 3; north wind.

The variables used were the burnt surface per resource configuration and the fire-extinction cost per resource configuration.

The most representative results belong to Classification 1. The biggest losses are produced when the wind direction is west-south west. When the wind direction is one of the other two, the fire was controlled much more in all the sampling units. The analysis of this classification (1) shows four main groups: Group 1, formed by the sample units where the fire is controlled and the resources cost is low; Group 2, Formed by the sample units where the burnt-surface value is low or

Table 1. Resources configuration

Configuration	Number of resources			
	Manual	Bulldozer	Fire pump	Fire-fighting plane
1	3	1	1	1
2	3	1	1	—
3	3	1	—	1
4	3	—	1	1
5	3	1	—	—
6	3	—	1	—
7	3	—	—	1
8	3	—	—	—
9	2	—	—	—
10	1	—	—	—

medium and the combat intensity is medium; Group 3, Formed by the sample units where the burnt-surface value is high in spite of all the combat resources. The fire wasn't controlled. Group 4, Formed by the sample units where all the resources are needed in order to maintain a low burnt-surface value.

Physical Parameters of Classification 1

Next, the influence of the physical parameters that have led to the former results was studied. An analysis of variance was carried out for the following variables: slope, accessibility, and types of fuel (Martín Fernández 1998). The aim of the analysis of variance is to check whether the mean values of the variables in the four groups mentioned above are similar or not. Fuel type is sorted according to the influence of the fuel in the fire-spread velocity (type A). This is an undimensional coefficient. It is obtained as the product of the opposition coefficient of the vegetation against the fire and the relative humidity of the fuel (Vélez 2000a). They are also sorted to its calorific power (type B). The results can be seen in Table 2.

Groups 1 and 2 are areas of low accessibility, but they are areas of pasture and shrubs where combat resources have good performance. However, Group 3 is a tree-covered area where the combat resources's performance is low. Fire cannot be controlled even if the accessibility is good. Finally, in Group 4 most of the areas have mild slopes and good accessibility. The combat resources can control the fire when they all work together because the main type of fuel is dense shrub.

Classification of the Remaining Points Applying Discriminate Analysis

The remaining points of the territory (see Figure 9) have been assigned to the former four groups. The Mahalanobis distance (Martín-Fernández 1998) has

Table 2. Physical parameters that characterize the groups of Classification 1

Group	Physical Parameter	Mean	S ₀
1	Accessibility	58.608 min	0.95 min
	Fuel-type A	0.2679	0.01
2	Accessibility	57.71 min	1.41 min
	Fuel-type A	0.27	0.01
3	Accessibility	30.347 min	1.48 min
	Fuel-type A	0.208	0.008
4	Accessibility	33.83 min	0.73 min
	Slope	13.7%	0.96%
	Fuel-type B	2.63 l/m ²	0.25 l/m ²

S = Standard deviation.

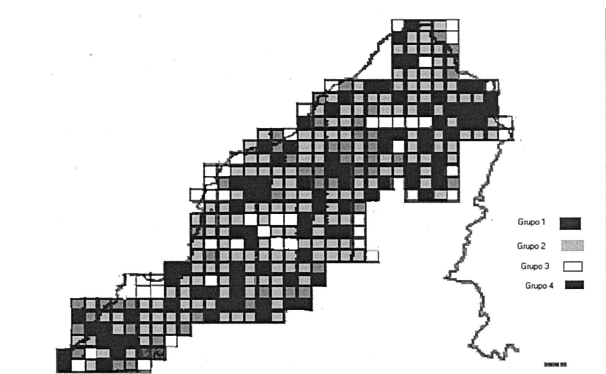


Figure 9. Classification of the territory according to the consequences of wildfires. In white, the areas of the highest losses where an efficient work of the available resources could not extinguish the fire (Group 3). In black the areas where all resources must be used to control the fire (Group 4). In grey, the areas with the least losses (Group 1 and 2).

been applied to make this classification. This is a non-Euclidean distance, applicable to quantitative variables, that measures the distance between the value of the variables between two points or between a point and a group of points.

$$d(i, j) = \sqrt{(w_i - w_j)^T C^{-1} (w_i - w_j)}$$

where: $w_i = (v_{i1}, \dots, v_{in})^T$ is the vector that represents measurements of the $(V_1, \dots, V_n)$ variables at point i ; $w_j = (v_{j1}, \dots, v_{jn})^T$ is the vector that represents measurements of the $(V_1, \dots, V_n)$ variables at point j ; and C is the covariance matrix for the $(V_1, \dots, V_n)$ variables.

—The variables used were slope, accessibility, and types of fuel. One point belongs to the group to which the Mahalanobis distance is the shortest

The result of the classification can be seen in Figure 9. Each point of the territory has been assigned to one of the four groups mentioned, once the smallest Ma-

halanobis distance from the point to the group is known. The total surface-area corresponding to each group are the following: Group 1, 192 km²; Group 2, 704 km²; Group 3, 192 km²; and Group 4, 380 km². As explained before, the points that belong to Group 3 are where the losses are the highest. A specific forestry management is necessary to avoid the risk of wildfires.

Conclusions

The optimization method for the management of the combat resources explained in this paper is a helpful tool for several aspects of wildfire management: fire-risk planning, simulation and training, and fire-fighting support.

—Fire-risk areas are identified through the simulation. Thus, planning the optimum allocation of the watchtowers to control the areas with more risk, and in the allocation of resource bases near the areas with more risk can be greatly improved.

In the 1998 study of the North west forests, the areas were identified where the consequences of fires were the worst. The accuracy of this study was further proved in the summer of 1999. There was a fire in the forest of San Lorenzo del Escorial, an area that belongs to Group 3, according to the classification made in 1998. This forest had recreative, protective, and productive uses. The global value of the losses were approximately 30 million dollars. It was the worst fire in the Madrid region that summer.

—Simulations can be carried out with different initial conditions, like different types of fuel, different number, and type of resources. The results of these simulations can help in the design of a preventive silviculture.

—The response time of the system is low enough that it can be applied to real combats. The system is updated during the combat with the real data, and then it calculates the best combat alternative. Thus, it is a helpful tool in the decision-making process.

Literature Cited

- Albini, F. 1986. Wildland fire spread by radiation, a model including fuel cooling by natural convection. *Combustion Science and Technology* 45:101–103.
- Anderson, H. D. 1989. A mathematical model for fire containment. *Canadian Journal of Forest Research* 19:997–1003.
- Andrews, P. L. 1986. BEHAVE: fire behavior predicting and fuel model system-burn subsystem. *General Technical Report. INT-194*. Intermountain research station Ogden, UT 84401. USDA Forest Service.
- Cerny, V. 1982. A thermodynamical approach to the travelling salesman problem and efficient simulation algorithm. *Preprint*. Institute of Physics and Biophysics. Comenius University, Bratislava.
- Dantzig, G. B. 1960. On the shortest route through a network. *Management Science* 6:177–187.
- Dupuy, I. L. 1991. Modélisation prédictive de la propagation des incendies de forêts. *Rapport Technique*. De la Université Claude Bernard Lyon 1. Inra, Station De Sylviculture Méditerranéenne, Avignon.
- Geman, S., D. Geman. 1984. Stochastic relaxation, Gibbs distributions and the Bayesian restoration of images *IEEE Pattern Analysis. PAMI* 6:721–741.
- ICONA. 1987. Guía fotográfica para identificación de modelos de combustibles. Ministerio de agricultura, pesca y alimentación. Madrid.
- Kirpatrick, S., S. D. Gellatt Jr, M. P. Vecchi. 1982. Optimization by simulated annealing. *IBM Internal Report*. Thomas J. Watson Research Center, Yorktown Heights, NY.
- Martín-Fernández, S. 1998. Aplicación de métodos bayesianos en el combate de incendios forestales. *Ph.D. Thesis*. Universidad Politécnica de Madrid.
- Martínez-Falero, E., A. Cazorla, J. Solana. 1995a. Scaling methods. Pages 193–235 in *Quantitative Techniques in Landscape Planning*. CRC-Lewis, New York.
- Martínez-Falero, E., M. Aguiló, E. Iglesias. 1995b. Landscape inventory. Pages 47–85 in *Quantitative Techniques in Landscape Planning*. CRC-Lewis, New York.
- Martínez-Millán, J., M. T. Alcaide. 1994. Utilización de CARDIN para la simulación de un incendio real. *Ecología* 8:3–11.
- Martínez-Millán, J., S. Saura. 2000. Simulación del comportamiento del fuego. El modelo CARDIN. Pages 9.20–9.35 in *La defensa contra incendios forestales: Fundamentos y experiencias*. McGraw Hill, Madrid.
- Mérida, J. C. 2000. Sistema BEHAVE. Pages 9.3–9.20 *La defensa contra incendios forestales: Fundamentos y experiencias*. McGraw Hill, Madrid.
- Mockus, J. 1989. Bayesian approach to global optimization. Kluwer, Dordrech.
- Rothermel, R. C., R. E. Burgan. 1984. BEHAVE: Fire behavior prediction and fuel modeling system. *General Technical Report. INT-167*. USDA Forest Service. Ogden, UT.
- Vélez, R. 2000a. Combustibles forestales: combustibilidad. Pages 7.1–7.16 in *La defensa contra incendios forestales: Fundamentos y experiencias*. McGraw Hill, Madrid.
- Vélez, R. 2000b. Perspectiva histórica de los incendios forestales en España. Pages 3.15–3.31 in *La defensa contra incendios forestales: Fundamentos y experiencias*. McGraw Hill, Madrid.
- Wybo, J. L. 1992. Knowledge-based analysis and real-time monitoring of spatial data. Application to forest fire prevention in French Riviera. In: *Proceedings of the International Emergency Management and Engineering Conference: Managing Risk with Computer Simulation*. Society of Computer Simulation, Orlando, FL.